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Repurposing a Depleted Gas Field for Underground Hydrogen Storage Using Surplus Wind Power

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Abstract

This study assesses the technical and commercial feasibility of repurposing a depleted gas reservoir for large-scale underground hydrogen storage (UHS) using surplus renewable electricity. Through the case study of the Hy1 field in the UK it is shown how legacy hydrocarbon infrastructure can be tailored to support net zero goals and updates similar strategies for hydrogen storage in regions such as the UAE.

The UK aims to cut net energy consumption by 30% by 2030 and achieve net zero by 2050. Surplus Scottish wind power is earmarked for green hydrogen production via electrolysis, with more than 1 GW of capacity planned. A detailed screening study identified the Hy1 onshore depleted gas field as a strong UHS candidate, featuring 18% porosity, 150 mD permeability, 50 m gross thickness, and 250 bar initial pressure. A 3D model was developed using CMG GEM¹ simulation which performs compositional simulation, combining seismic data and 15 years of production history. Multiple well placements and scenarios (H₂, CO₂, N₂) were tested for scientific and economic performance. Simulation results confirmed that the Hy1 field behaves as a volumetric tank, with an estimated gas-in-place volume of 8 Bscf, validating its suitability for UHS. Crestal wells located within the anticline structure displayed superior injection and withdrawal efficiency compared to other configurations. Hydrogen was identified as the most cost-effective cushion gas due to its availability on-site from electrolysis and favorable performance characteristics. The field demonstrated the capacity to store up to 15,000 tonnes of hydrogen, with achievable injection and withdrawal rates of 300 and 200 tonnes per day, respectively. Major risks include trap integrity, caprock sealing effectiveness, microbial activity that may alter stored hydrogen, and the co-production of residual methane. These risks require management through optimized cushion gas selection and the installation of surface separation systems to ensure hydrogen purity. Clustering multiple UHS sites was also shown to be essential for scaling up to gigawatt-level electrolyzer deployments, ensuring a dispatchable and economically viable hydrogen supply chain.

This study is among the first to simulate cyclic hydrogen injection in a mature UK gas field while integrating economic assessments of different cushion gas options. It presents a practical and transferable workflow for other regions, particularly the UAE, where depleted fields such as Habshan and Bab could be

repurposed using solar-powered electrolysis and extensive legacy subsurface data. This framework enables faster screening, securing, and development of UHS projects supporting both domestic decarbonization and hydrogen export strategies. Furthermore, this study introduces a new operational consideration management of co-produced methane during withdrawal which is expected to become a significant economic and regulatory factor in future UHS developments.

Introduction

The burning of fossil fuels and the increased emission of greenhouse gases into the atmosphere have adversely contributed to global warming. Therefore, energy systems have been seeking renewable sources, including wind, solar, and hydropower.² As countries accelerate their decarbonization efforts, hydrogen is emerging as a fundamental part of the transition from fossil fuels to a cleaner and more adaptable energy system. While renewable sources like wind and solar are essential to this shift, their power generation is intermittent due to weather and seasonal changes. This makes it difficult to. [Figure 1](#) shows how the UK's electricity generation mix has changed over the past year, highlighting the growing share of renewables alongside fossil fuels, nuclear, and interconnectors. However, to fully rely on renewables, we need technologies that can accommodate their fluctuations. Hydrogen helps addressing this by acting as both a clean fuel and a method to store excess renewable energy. This flexibility makes hydrogen a vital mechanism for balancing supply and demand and supporting the reliability of future energy systems.

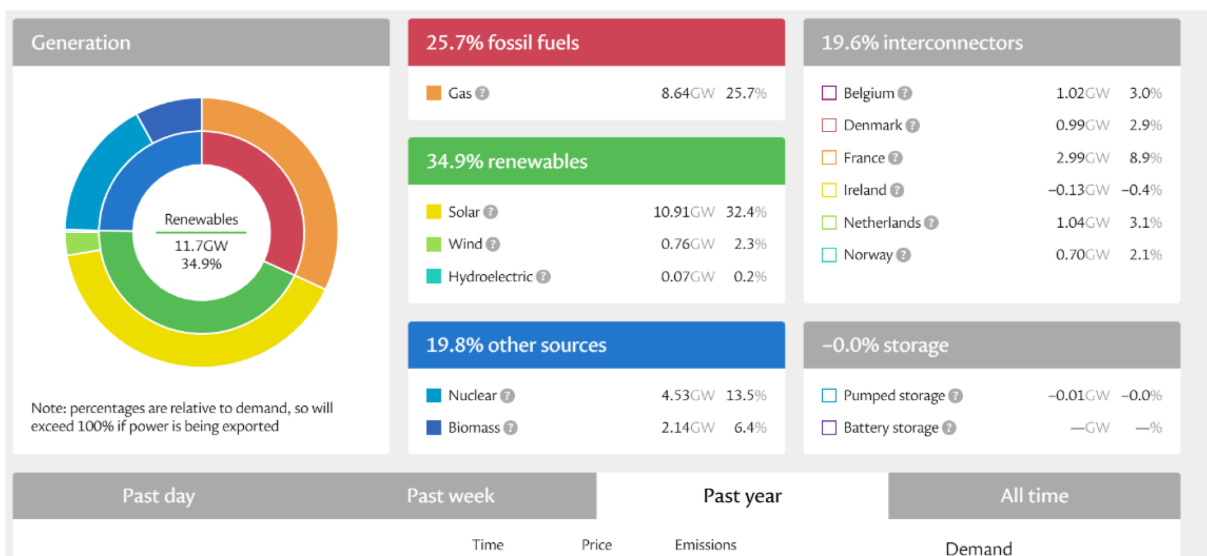


Figure 1—Electricity generation mix in the UK, highlighting the contribution of renewables, fossil fuels, nuclear, and interconnectors over the past year.³

To express hydrogen's potential as a fundamental energy carrier, robust and extensive storage solutions are critical. Among the available options, underground hydrogen storage (UHS) stands out as the most practical method for accommodating bulk quantities of hydrogen over long periods. Leveraging decades of operational experience from underground natural gas storage, depleted gas fields offer particularly promising sites for UHS. Their established infrastructure demonstrated ability to contain gases securely, and substantial pore volumes make them well-suited for storing hydrogen on a large scale and supporting a stable, flexible energy system. As shown in [Figure 2](#), many such formations across the UK have already been characterized for underground gas storage potential, including the Hy1 field studied in this work.

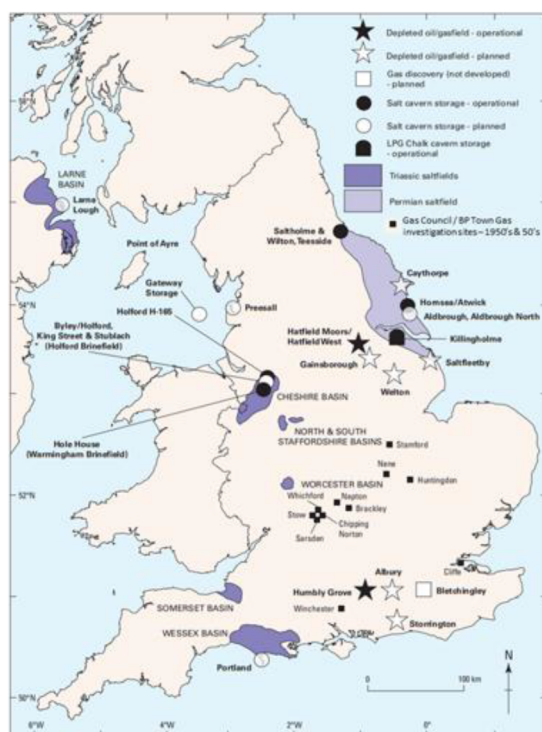


Figure 2—Source: BGS D.J. EVANS. 2007. An appraisal of Underground Gas Storage technologies⁴

Even though depleted gas fields have great potential for storing hydrogen, they are not used as much as salt caverns currently. Hence, there are still some uncertainties as to how hydrogen flows through the rocks, whether it reacts with the reservoir materials, if microbes might affect it, and whether the caprock can hold it safely for a long time. These unknowns have made companies hesitant to use these fields. This study tries to bridge those gaps by looking at the Hy1 depleted gas field in the UK and seeing if it could safely store green hydrogen. We used detailed reservoir data and ran advanced simulations to check if this kind of storage is possible and efficient. Figure 3 shows how efficient this process is from making hydrogen through electrolysis to using it later for power generation during peak demand. The primary aim is to see how well hydrogen can be injected and taken out in cycles, specifically using extra wind energy. We also reviewed at how to reduce the amount of cushion gas needed, tested different well setups, and studied possible risks like gas mixing or microbes affecting with the hydrogen. The purpose of cushion gas is to maintain the required pressure and transfer rate and undergo compression and expansion during alternating injection and expansion periods, respectively.² Additionally, the efficacy of subsurface hydrogen storage can be improved by water injection, with peak hydrogen recovery observed when methane is used as the support gas. Ultimately, we took the findings and applied them to fields in the UAE (like Bab and Habshan), where conditions for producing and storing green hydrogen are favourable.

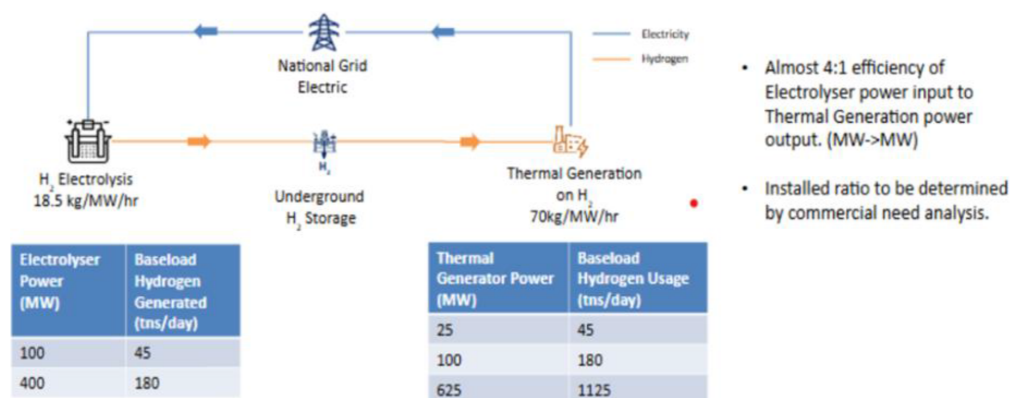


Figure 3—Electrolysis-to-Thermal Generation Efficiency for Underground Hydrogen Storage in Power Peaking Applications

Methodology

Site Selection Criteria

The initial phase of this hydrogen storage feasibility study involved a broad screening of ten subsurface sites between depleted gas and aquifers across the UK. Following this evaluation, the Hy1 field was selected as the most promising candidate based on a combination of favourable geological and operational characteristics. Structurally, Hy1 is defined by a well-developed anticline trap, which provides a reliable seal for gas containment. Its selection was also supported by the availability of rich subsurface data including seismic surveys, well logs, and production history allowing for accurate reservoir characterization and robust dynamic modelling. From a reservoir quality standpoint, Hy1 presents favourable properties, with an average porosity of 18% and permeability of around 150 millidarcies. These values support both adequate storage capacity and the feasibility of cyclic hydrogen injection and withdrawal. Additionally, the reservoir depth and initial pressure of approximately 250 bar help maintain sufficient gas density, which reduces the threat of leakage and enhances overall storage efficiency.

To provide broader context, [Table 1](#) presents a comparative overview of UHS feasibility across the other screened UK sites. This comparison highlights Hy1's relative advantages and supports its selection for detailed modelling and evaluation in this study.

Table 1—Comparative Overview of our key finding in Underground Hydrogen Storage Feasibility Studies Across Selected UK Subsurface Sites. (Gupta,S. 2025. Study of Hydrogen. Internal Report, Rara Consultancy Energy, Abu Dhabi Dubai (Unpublished))

Site Name	Field Type	Work Performed	Key area of study	Simulation Model
Hy1	Depleted Gas	Full UHS feasibility study completed, layered simulation model developed	Cushion gas requirement, well integrity, reservoir performance, reuse of existing infrastructure	GEM compositional model created, Basic well model in Coflow
Aquifer	Aquifer	UHS feasibility study completed	Reservoir characterization, seal integrity, injectivity, hydrogen migration, trapping mechanisms	GEM compositional model created
Field B	Depleted Gas	Limited study: Storage capacity estimation and basic history match due to lack of detailed production and pressure data	Volumetric capacity assessment, preliminary dynamic behavior using limited data	GEM compositional model (basic setup)
Field C	Depleted Gas with Aquifer	Hydrogen storage simulation using GEM, No production data available	Aquifer response, upconing risk, pressure propagation, learning from dynamic behavior	GEM compositional model created

Model Development

After selecting the Hy1 field, a comprehensive dynamic reservoir model was developed using the CMG GEM¹ simulator to evaluate its potential for UHS. The model was calibrated through history matching, using 15 years of actual production and pressure data from the field. This calibration ensured the model could reliably reproduce historical reservoir behaviour and be used confidently to forecast future storage scenarios. To guide this evaluation, Figure 4 presents the methodological framework followed in this study from site screening and data integration to dynamic simulation and performance analysis. This framework helped streamline the workflow and ensure all critical aspects of hydrogen storage assessment were systematically addressed.

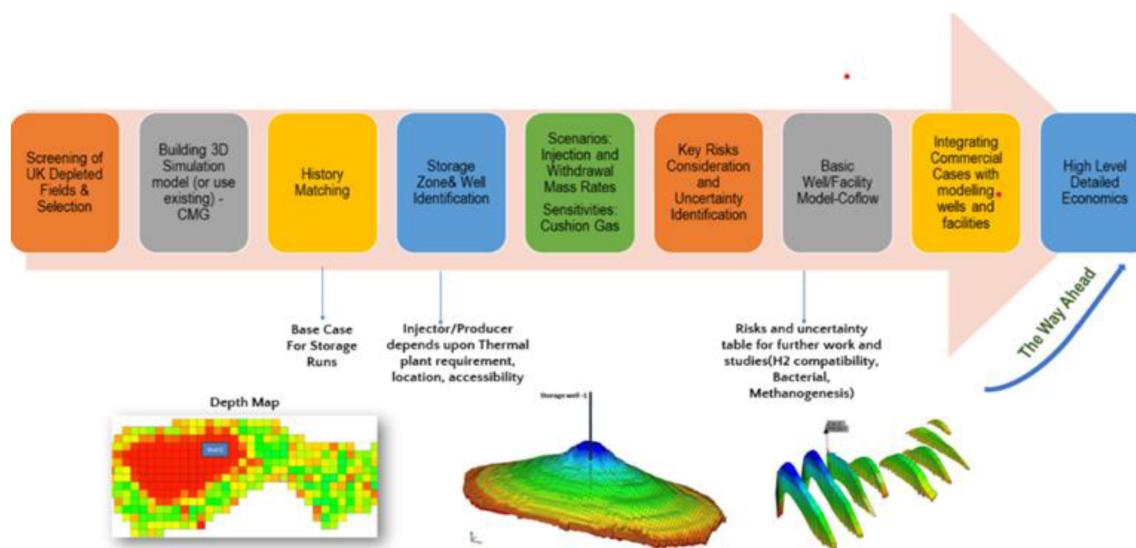


Figure 4—Methodological Framework for Evaluating Hy1 Depleted Gas Field for Hydrogen Storage¹

In the next phase, a series of sensitivity analyses were performed to assess how uncertainty in key reservoir properties may alter hydrogen storage outcomes. Parameters such as porosity, permeability, rock compressibility 2×10^{-6} psia and relative permeability endpoints were varied to test the model's response. These analyses helped measure the impact of subsurface variability on injectivity, storage capacity, and withdrawal rates, overall validating the model's robustness across a range of different scenarios. To validate this, Figure 5 shows the relationship between porosity and rock compressibility across various lithologies, including consolidated sandstones, friable formations, and limestones highlighting how lithological differences can directly impact hydrogen storage behaviour.

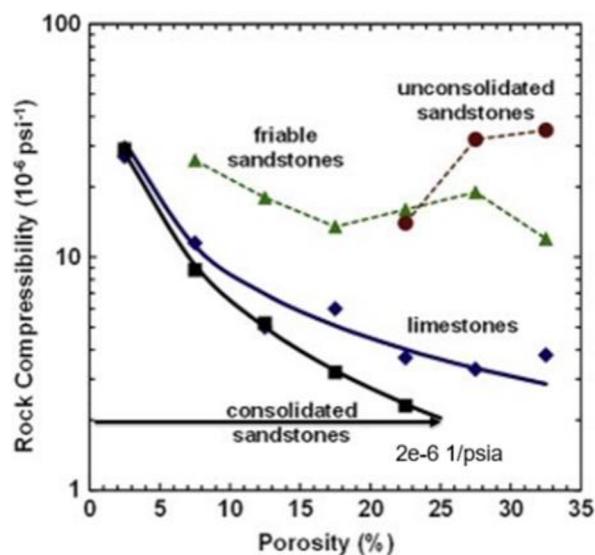


Figure 5—Relationship between porosity and rock compressibility for various lithologies⁵

The simulation results provided useful information as to how hydrogen behaves under real conditions. One major finding was that effective pressure management is critical. Optimal performance was attained when injection and withdrawal cycles were kept within a defined pressure window, which helped maintain hydrogen density while avoiding excessive stress on the caprock. The model also demonstrated that well placement and injection strategy play an important role in performance. Poorly placed wells or uneven injection could lead to early hydrogen breakthrough or inefficient use of the reservoir volume. These results highlight the importance of tailoring operational plans to the specific characteristics of each field. Taking these factors into consideration prior in the planning phase can ensure that storage systems are both efficient and safe, ultimately making the move from concept to field deployment more direct.

Results and Discussion

History Match and Base Case Validation

The dynamic simulation model developed for the Hy1 field showed solid agreement with historical reservoir pressure trends, demonstrating its reliability for future hydrogen storage predictions. A history matching process was carried out using 15 years of actual production and pressure data, allowing the model to precisely mimic the field's past behaviour. This action was essential to make certain the model could be trusted for forecasting cyclic hydrogen injection and withdrawal performance. The reservoir initially held a pressure of approximately 400 bar, and the simulation successfully captured the pressure decline down to an abandonment level of 40 bar. As shown in Figure 6, the simulated pressure profile aligns closely with historical measurements over a 20-year period, providing confidence in the model's ability to represent the reservoir dynamics accurately and support forward-looking hydrogen storage scenarios.



Figure 6—Comparison between historical pressure data and simulated pressure profile for the Hy1 field. The model accurately reflects 20 years of reservoir behavior, confirming its reliability for forward simulation. (Gupta,S. 2025. Study of Hydrogen. Internal Report, Rara Consultancy Energy, Abu Dhabi Dubai (Unpublished))

Permeability Characterization and Sensitivity

To simulate realistic flow behaviour within the Hy1 reservoir, a layer cake geological model was constructed to capture the field's vertical heterogeneity. This approach allowed for a simplified yet geologically representative layering of the reservoir, which is essential when modelling fluid movement and pressure changes across zones of varying quality. A constant permeability of 300 millidarcies (mD) was assigned to the primary productive layers (layers 6, 7, 8, and 9), as informed by available well log interpretations. These layers were known to contribute the most to historical gas production and were therefore modelled to reflect higher flow capacity. However, drill stem test (DST) reports pointed to significantly lower permeability values (~14 mD) in the upper reservoir zones, revealing clear vertical contrast in flow properties. To account for this, and in the absence of DST pressure-time data for deeper validation, a series of sensitivity analyses were conducted to test the impact of varying permeability distributions on both the history match quality and overall storage performance.

As illustrated in Figure 7, the cross-sectional view and accompanying histogram highlight how permeability was distributed across the model. While the productive zones were assigned fixed values, alternate scenarios were run using different permeability contrasts in non-productive layers to observe changes in reservoir pressure behaviour. This analysis helped determine a realistic permeability range that maintains volumetric consistency and honours observed production while not compromising the physical behaviour of hydrogen during injection and withdrawal.

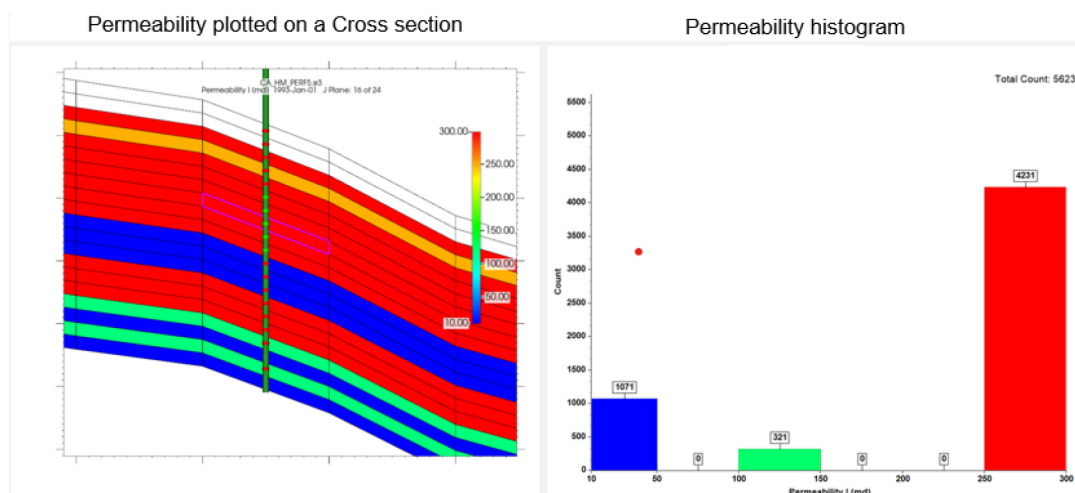


Figure 7—(a) Cross-sectional view and (b) histogram showing permeability distribution in the Hy1 reservoir model. Constant permeability was assigned to productive zones, while sensitivity runs helped understand its impact on pressure behaviour.¹

Hydrogen Storage Performance

With the reservoir model successfully validated, it was then used to simulate hydrogen storage performance under cyclic injection and withdrawal scenarios. The Hy1 field demonstrated the capacity to store up to 15,000 tonnes of hydrogen, with operational injection and withdrawal rates modelled at 300 tonnes per day (tpd) and 200 tpd, respectively. These values are within a feasible operational range for grid-scale storage and energy buffering applications. Simulation results revealed that crestal well placements positioned near the top of the structure provided the best injectivity and productivity due to higher gas mobility and lower water saturation in those zones. Interestingly, the model showed that higher injection rates shortened cycle times but also reduced storage efficiency, likely because gas bypassed certain low-permeability regions, leading to uneven pore space usage. Conversely, lower injection rates allowed for better gas distribution, more complete pore filling, and improved containment, albeit at the cost of longer cycle times. As shown in Figure 8, the 3D view of the reservoir highlights the geometry of the Hy1 field alongside simulated hydrogen injection and withdrawal cycles. The plots illustrate how storage volume and flow rates vary over time, offering key insights into how operational strategies affect both capacity and performance. These results emphasize the importance of balancing rate-driven operational efficiency with reservoir utilization to achieve long-term sustainability. Optimizing injection schedules may allow operators to tailor performance to seasonal energy demands. The Hy1 field thus presents a viable and flexible option for large-scale green hydrogen storage in a practical setting. These findings also serve as a reference point for scaling up hydrogen storage in other depleted fields with similar geological characteristics. Further modelling that includes economic constraints and infrastructure combination will be essential to support commercial circulation.

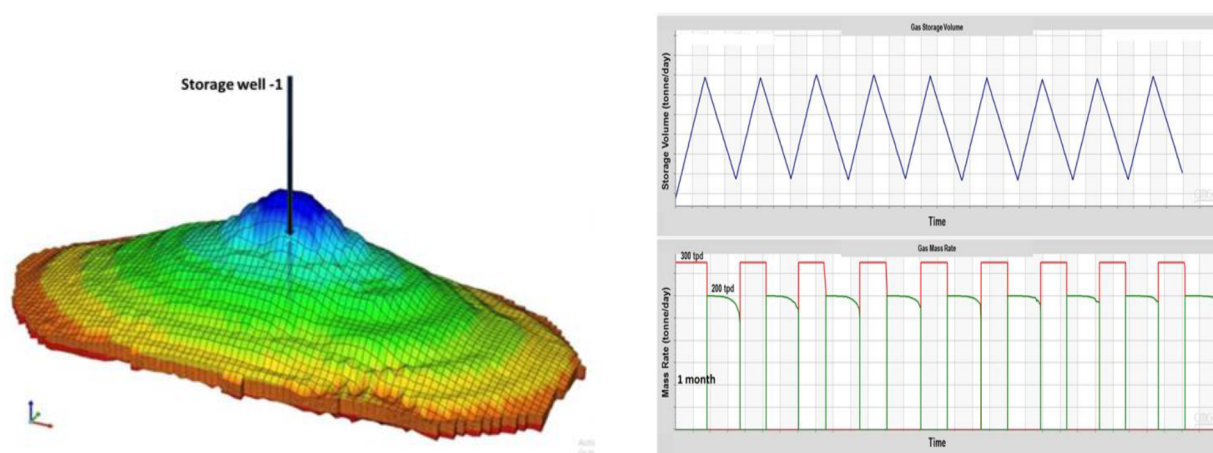


Figure 8—3D view of reservoir geometry and simulation output for hydrogen storage cycles in the Hy1 field. Plots show injection/withdrawal rates and gas storage volume changes over time.¹

Surface Facility Considerations

An important observation was the co-production of residual methane during hydrogen withdrawal. This outcome comes from the Hy1 field's previous use as a natural gas reservoir and introduces both analytical and regulatory challenges. To maintain hydrogen purity for downstream use, surface separation facilities are necessary to isolate methane. Moreover, the presence of hydrocarbons, even as a by-product, could trigger licensing complications, potentially requiring a hydrocarbon production license in addition to a hydrogen storage permit. This could affect the permitting timeline, raise compliance costs, and create regulatory uncertainty. These considerations are critical for project developers and policymakers when testing commercial viability.

Uncertainty from Aquifer Influence

During the study, one of the important unknowns was how water present beneath or next to the gas field, an aquifer may affect hydrogen storage practices. Although there was some history of water being produced from one of the wells (Well-1), we do not have enough detailed information to fully understand the size, connectivity, or behaviour of this aquifer system. This lack of knowledge creates uncertainty in how the aquifer could influence pressure within the reservoir during hydrogen injection and withdrawal cycles. For example, the aquifer might help by providing pressure support, which could be beneficial, or it could cause unwanted movement of fluids that rises the risk of leaks or reduces storage efficiency.

Since we cannot precisely predict how the aquifer will behave, especially under different storage scenarios over time, this uncertainty was identified as a risk factor for the project. Handling reservoir pressure safely is important to avoid damage to the caprock seal and to prevent the loss of stored hydrogen. As a result, we have included aquifer influence in the risk register, emphasizing the need for further detailed studies and sensitivity analyses. These future investigations will help clarify whether the aquifer will support or compromise the reservoir's pressure system and overall storage effectiveness, enabling better risk management and safer, more reliable underground hydrogen storage operations.

Discussion

Dynamic Reservoir Simulation Advantage

Dynamic reservoir simulation proved to be more effective than simple static screening methods. It helped capture how hydrogen realistically flows through the reservoir, how pressure changes over time, and how well the reservoir handles repeated injection and withdrawal cycles. The model was based on real physical reservoir properties, which means the simulation results were grounded in actual geological conditions. This

approach offers a reliable and cost-efficient way to reduce uncertainty in underground hydrogen storage (UHS) projects. By using this method, operators can better evaluate and compare different sites early on before spending money on detailed field testing or drilling.

Geological Integrity and Feature Calibration

One of the main strengths of this study is that it was based on real, field-specific geological data rather than assumptions. Properties such as porosity, permeability, and caprock integrity were taken directly from the Hy1 field dataset. This made it possible to build a simulation model that closely reflects actual reservoir conditions. Accurate input data is especially important for underground hydrogen storage (UHS), where even small errors in rock properties can lead to incorrect predictions about storage capacity, injectivity, or the risk of leakage. In addition, the study incorporated well-established relationships between rock type and compressibility to further validate the model's behaviour under pressure. By grounding the model in real reservoir characteristics and established rock mechanics, the study offers a reliable and transferable approach that can be applied to other depleted fields such as those in the UAE when similar data is available. This data-driven method helps ensure that performance forecasts are realistic and increases confidence in the safety and efficiency of proposed hydrogen storage projects. It also supports informed decision-making by operators and regulators as they assess the potential for reusing mature gas fields for clean energy applications.

Implementation in UAE

The simulation-based approach developed in this research offers a practical and cost-effective way to speed up underground hydrogen storage (UHS) projects. By allowing quick, physics-based screening of potential sites, operators can identify and prioritize fields like Hy1 that have high storage capacity, strong geological traps, and good well configurations. This reduces the need for expensive field configuration and helps fast-track the selection of the best sites for development. This method acts as a catalyst for decision-making, which is important in regions aiming for rapid decarbonization and large-scale renewable combination.

The approach is especially useful in the UAE, where several advantages make it easy to apply. First, the country has strong solar energy potential, which supports low-cost hydrogen production through electrolysis. Second, decades of oil and gas operations have created a large amount of high-quality subsurface data. Third, mature gas fields like Habshan and Bab already have the infrastructure needed for hydrogen storage and can be repurposed. These factors make it possible for the UAE to move quickly from technical evaluation to actual hydrogen storage operations. By reusing existing resources, this strategy helps support national energy transition goals while also positioning the UAE as a global leader in green hydrogen production and export.

Limitations and Future Scope

While this study shows a strong example of how a depleted UK gas field like Hy1 can be repurposed for underground hydrogen storage (UHS), it is important to acknowledge its current limitations. The analysis focuses on a single geological setting, which means the findings may not directly apply to other types of reservoirs. To make the workflow more widely useful, future research should test it across a wider range of subsurface conditions such as different rock types, structural traps, and aquifer interactions. In addition, some important processes were either simplified or not included in the current simulation framework. These need to be conveyed in future work to improve the accuracy and reliability of storage performance predictions:

1. **Geomechanical Effects:** Repeated hydrogen injection and withdrawal cycles can change pressure conditions in the reservoir and caprock, potentially causing stress, deformation, or even fractures. Including geomechanical models will help assess how the rock behaves under operational conditions and ensures long-term storage safety.

2. **Microbial and Biogeochemical Risks:** Microbial activity like methanogenesis can change the composition of stored hydrogen and affect how much can be recovered. To better understand these risks, future studies should include coupled biogeochemical and reservoir simulations and explore ways to prevent or manage microbial effects.
3. **Economic Optimization:** Just because hydrogen storage is technically possible doesn't mean it's economically viable. Future work should link reservoir simulations with cost models for large-scale electrolyzers and hydrogen markets. This would help optimize operating strategies to balance cost, efficiency, and grid reliability.
4. **Probabilistic Modelling:** Finally, adding probabilistic or uncertainty-based approaches can provide a clearer picture of risks. By quantifying the range of possible outcomes for storage capacity, injectivity, and aquifer response, project planners can make more informed decisions and better manage risk.

Conclusions

This study ensures the feasibility of repurposing the Hy1 depleted gas field for large-scale UHS. We have obtained that the field can store up to 15,000 tonnes of hydrogen, with injection and withdrawal rates of 300 and 200 tonnes per day, supporting grid-scale energy needs. The simulation model, built using 15 years of production and pressure data, provides strong confidence in performance under cyclic operations. Crestal well placement and using hydrogen from nearby electrolyzers as cushion gas improved both injectivity and withdrawal efficiency, making the system suitable for balancing intermittent renewable power.

Major risks such as caprock integrity, aquifer interaction, and co-production of residual methane were identified. These require ongoing monitoring and surface separation systems to maintain hydrogen purity and ensure safe storage. The model's close alignment with real geological properties improves trust in its predictions and supports regulatory compliance. Underground gas storage provides a promising approach to manage the seasonal variability of hydrogen production and demand.²

The developed workflow is practical, cost-effective, and applicable to other regions. In particular, the UAE with mature gas fields like Bab and Habshan, strong solar potential, and detailed subsurface data is well-positioned to apply this approach. By combining legacy infrastructure with green hydrogen production, the UAE can advance its decarbonization goals and strengthen its role in the global hydrogen market.

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